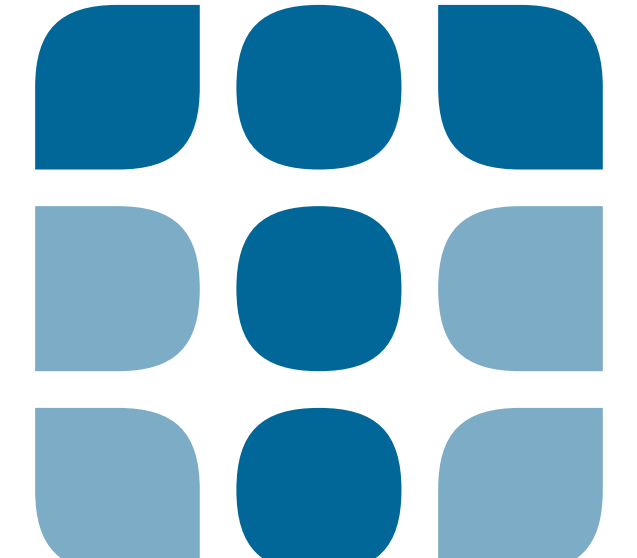


Multi-Band Patch Antenna Array for Out-of-Band Aided Millimeter Wave Communication



Faruk Pasic*, Jure Soklič†, Robert Langwieser*,
Stefan Schwarz* and Christoph F. Mecklenbräuer*



*Institute of Telecommunications, TU Wien, Vienna, Austria

†PIDSO Propagation Ideas and Solutions GmbH, Vienna, Austria

IEEE International Workshop on Antenna Technology (iWAT 2026)

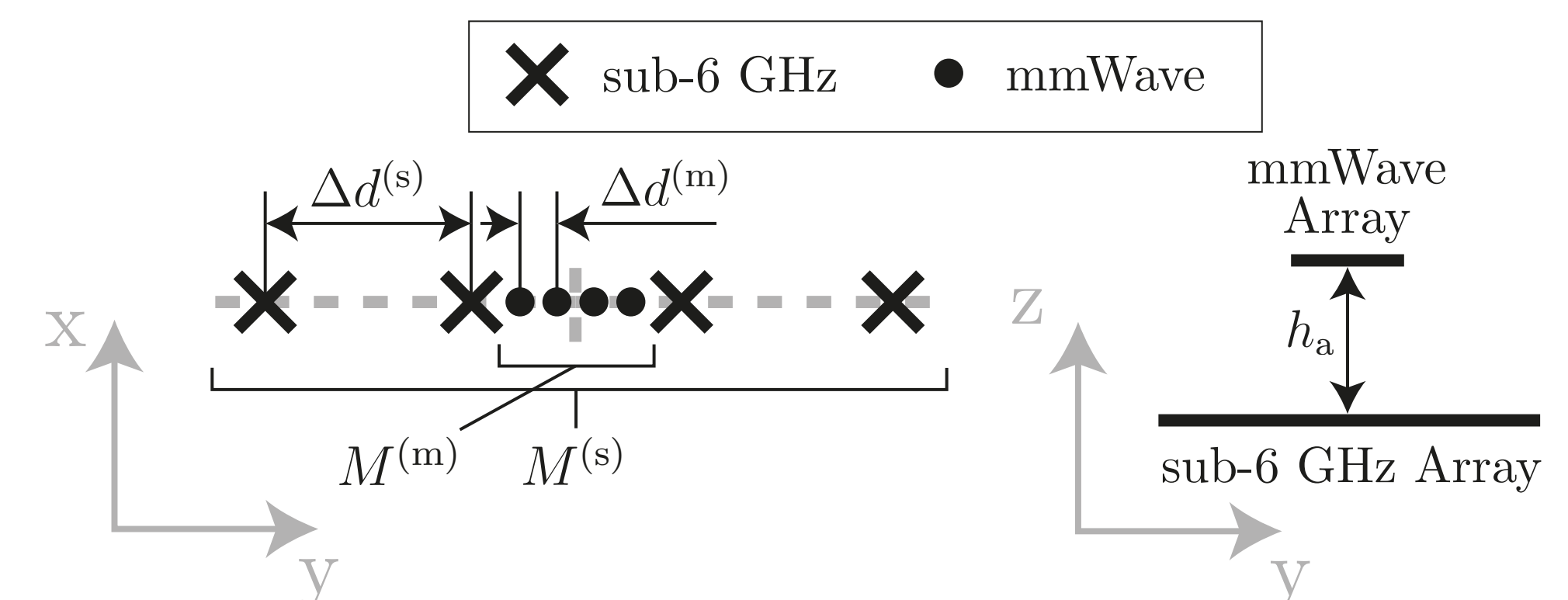
PIDSO

① Motivation

- ◇ mmWave bands (24 GHz – 300 GHz)
 - High data rates with challenging propagation
 - Reliable sub-6 GHz out-of-band information
 - Co-located and aligned arrays [1], [2]
 - Similar multi-path propagation
- ◇ **Problem:** Challenging array design
 - Impact on radiation pattern and performance

② Multi-Band Antenna Array Design

- ◇ Co-located and aligned sub-6 GHz and mmWave arrays (sharing the same geometric center)
- ◇ Uniform linear arrays (ULAs) of patch elements with half-wavelength antenna spacing
- ◇ Arrays separated by an inter-array spacing of h_a
- ◇ Investigate realized gain patterns of a sub-6 GHz patch element or sub-6 GHz antenna array

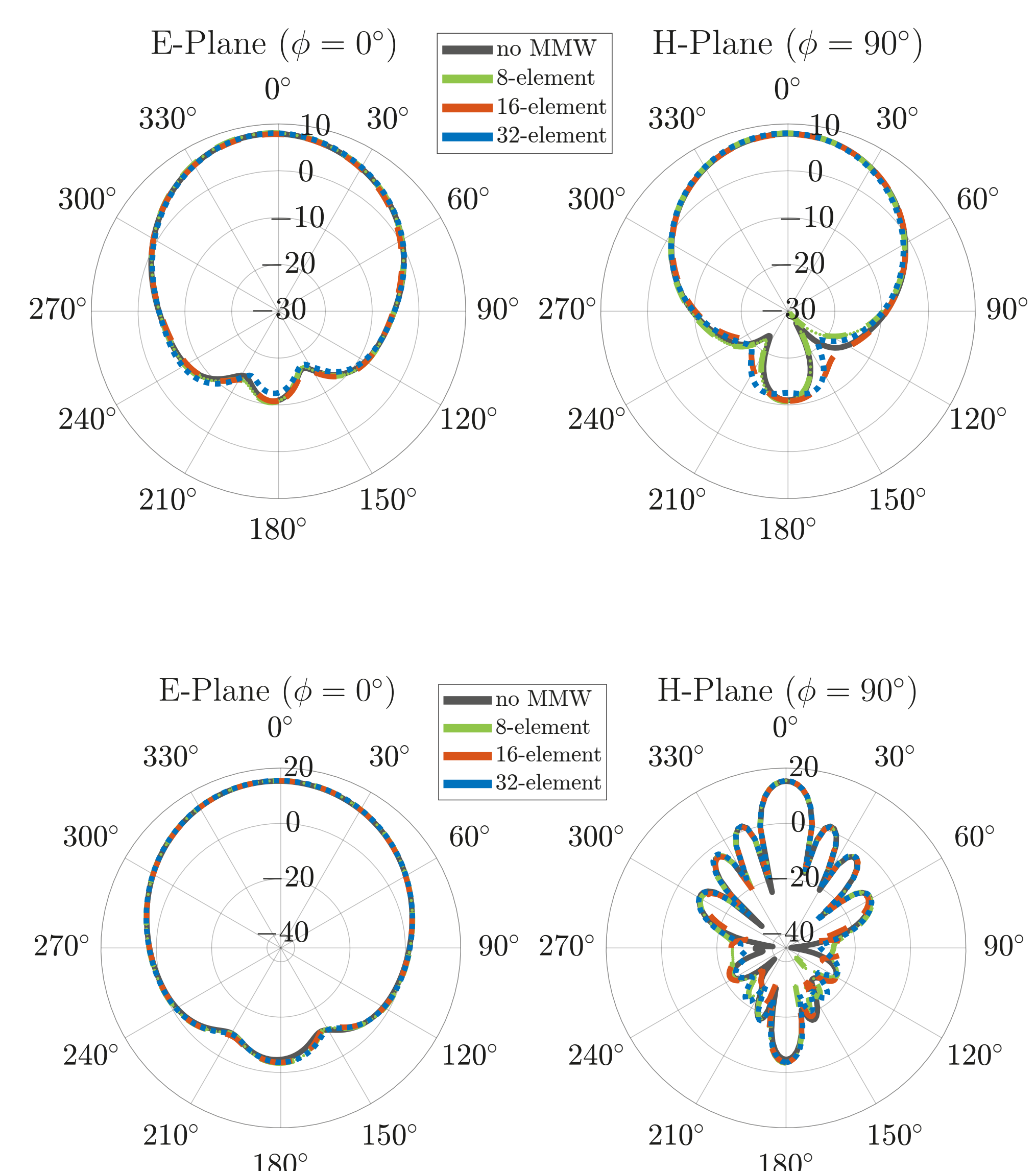
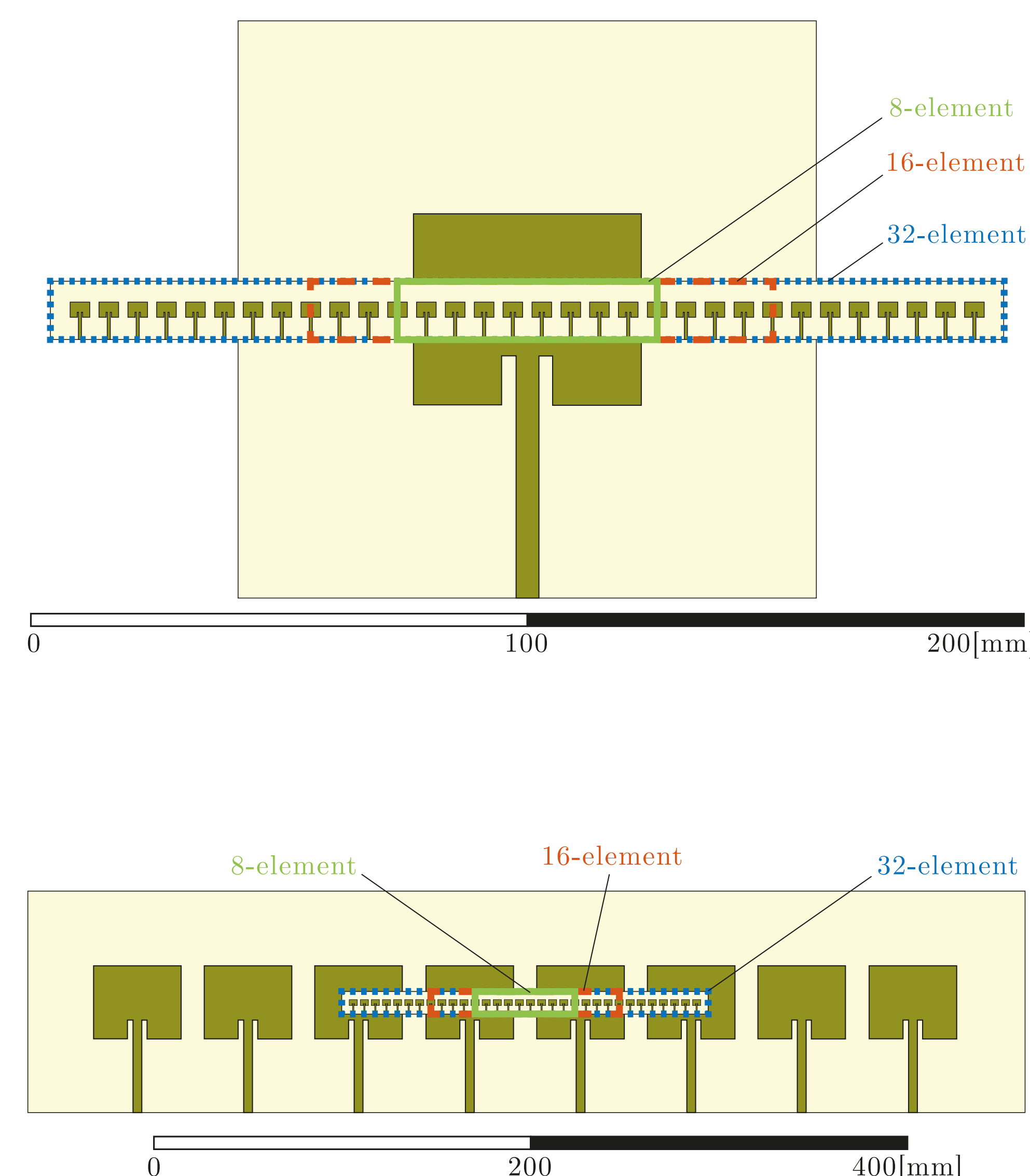


③ Parameters & Procedure

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Substrate Type	RT-Duroid 5880	RO4350B
Relative Dielectric Constant ϵ_r	2.2	3.66
Substrate Thickness h_s	1.5 mm	0.254 mm
Copper Thickness h_c	70 μ m	70 μ m
Dissipation Factor $\tan \delta$	0.0004	0.0037
Microstrip Line Width w_m	4.53307 mm	0.504685 mm
Patch Element Width w_p	46.47 mm	3.851 mm
Patch Element Length l_p	38.89 mm	2.985 mm
Inset Feed Cut Width w_{cut}	5.7899 mm	0.5276 mm
Inset Feed Cut Length l_{cut}	10 mm	1.005 mm

- ◇ Different mmWave arrays placed in front with an inter-array spacing of $h_a = 10$ mm
- ◇ Standalone sub-6 GHz element/array without any mmWave structure used as a baseline reference
- ◇ Simulations performed using ANSYS HFSS
- ◇ Measurements in the anechoic chamber at TU Wien

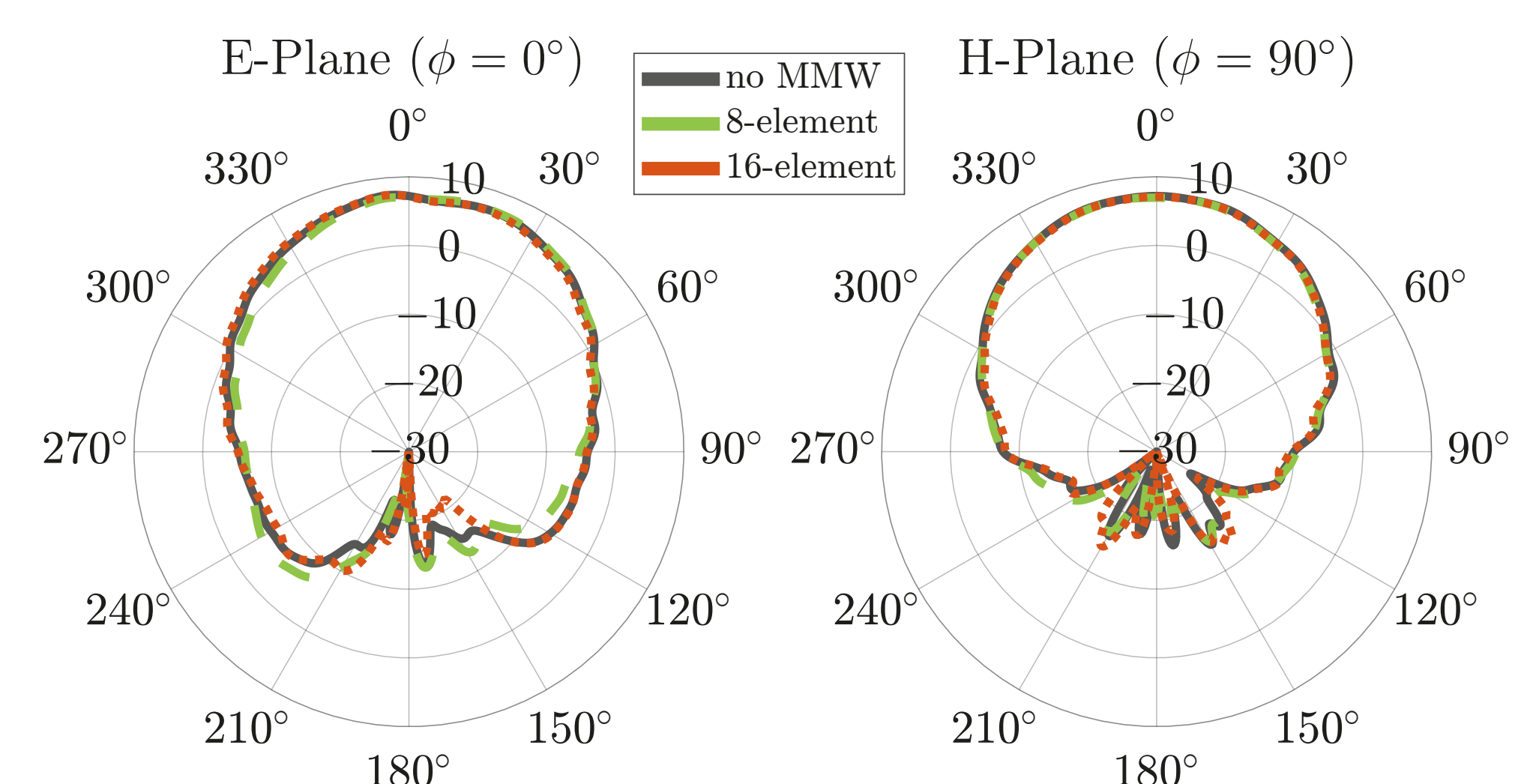
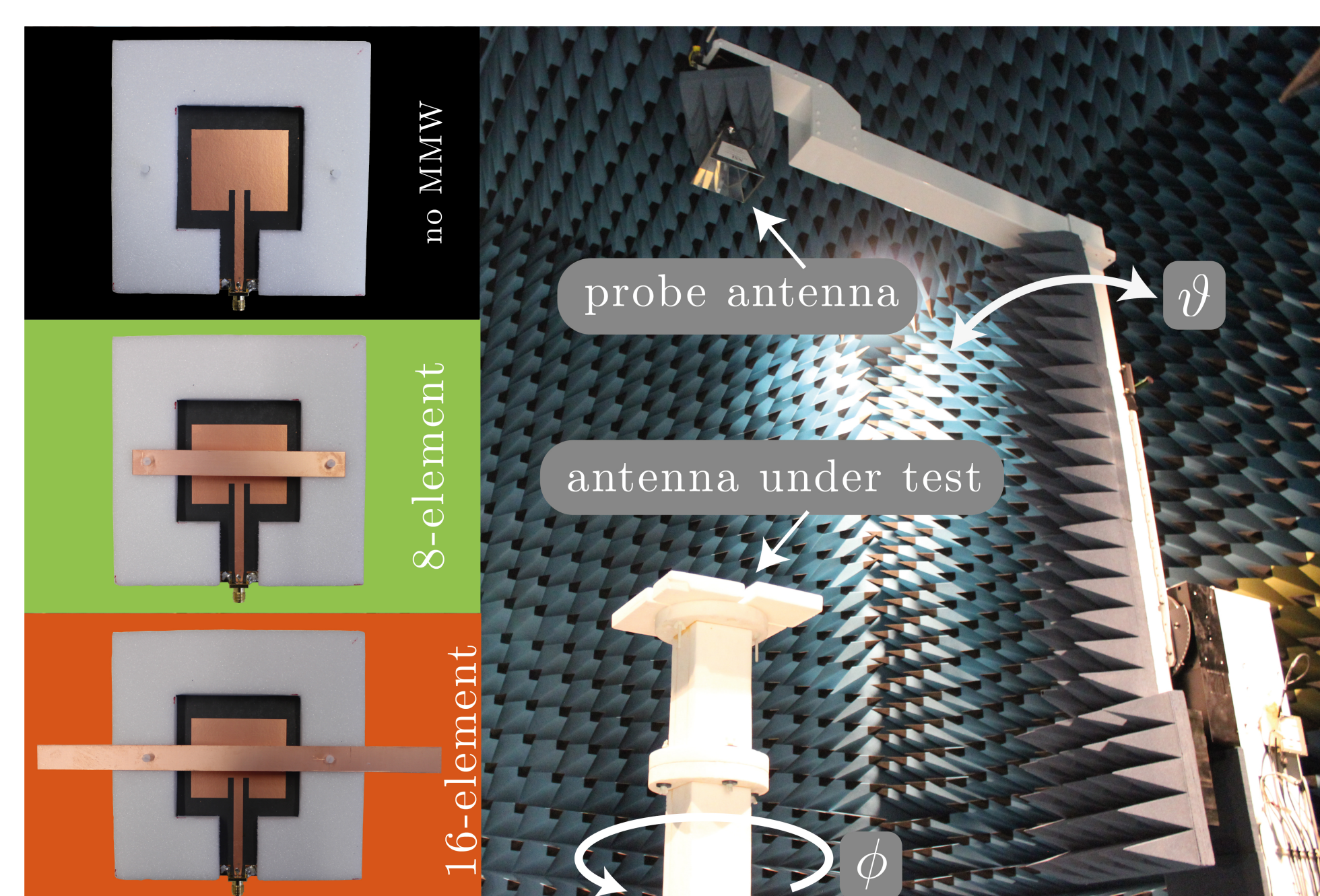
④ Analysis using Simulations



⑥ Results

		mmWave Array Configuration	Realized Gain	HPBW	
				E-plane	H-plane
Single Antenna	Simulated	no mmWave	7.96 dBi	68°	64°
		8-element	7.86 dBi	68°	62°
		16-element	7.84 dBi	68°	64°
		32-element	7.86 dBi	68°	62°
	Measured	no mmWave	7.20 dBi	75°	74°
		8-element	6.95 dBi	71°	74°
Array	Simulated	no mmWave	15.33 dBi	68°	12°
		8-element	15.23 dBi	68°	12°
		16-element	15.20 dBi	66°	12°
		32-element	15.18 dBi	68°	12°

⑤ Analysis using Measurements



⑦ Summary

- ◇ Presence of the mmWave array has only a minor impact on the radiation characteristics of both a single sub-6 GHz patch element and sub-6 GHz ULA
- ◇ Co-located sub-6 GHz and mmWave ULA configurations are feasible and introduce only negligible performance degradation
- ◇ Future work should extend this analysis to uniform planar array (UPA) configurations to assess radiation effects in two-dimensional antenna structures

⑧ References

- [1] A. Ali et al., "Spatial Covariance Estimation for mmWave Hybrid Systems Using Out-of-Band Information," IEEE Transactions on Wireless Communications, 2019
- [2] F. Pasic et al., "Millimeter Wave MIMO Channel Estimation Using Sub-6 GHz Out-of-Band Information," IEEE Transactions on Communications, 2025